

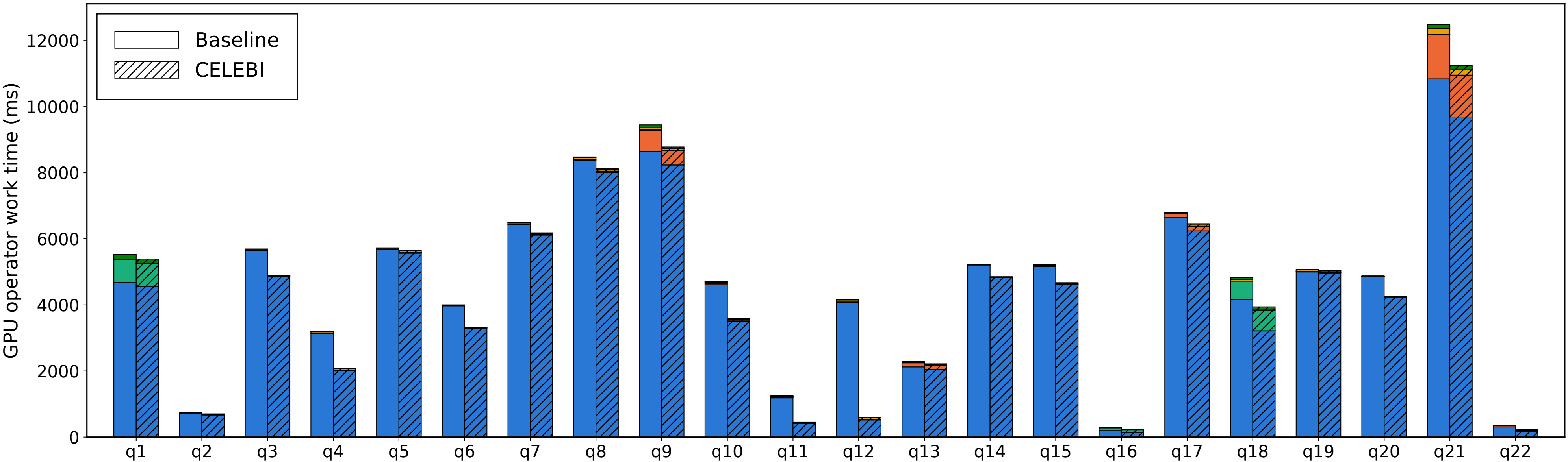


Figure 6: Total latency breakdown by operator type, Baseline vs. CELEBI (SF50, 4GB cache budget). The baseline query order triggers real page eviction (2 sweeps, 792MB evicted) that reordering avoids entirely, reducing total GPU operator time by 13.1% (106.8s -> 92.9s).